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Gene flow and genetic drift in urban environments

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Abstract

Evidence is growing that human modification of landscapes has dramatically altered evolutionary processes. In urban population genetic studies, urbanization is typically predicted to act as a barrier that isolates populations of species, leading to increased genetic drift within populations and reduced gene flow between populations. However, urbanization may also facilitate dispersal among populations, leading to higher genetic diversity within, and lower differentiation between, urban populations. We reviewed the literature on nonadaptive urban evolution to evaluate the support for each of these urban fragmentation and facilitation models. In a review of the literature with supporting quantitative analyses of 167 published urban population genetics studies, we found a weak signature of reduced within-population genetic diversity and no evidence of consistently increased between-population genetic differentiation associated with urbanization. In addition, we found that urban landscape features act as barriers or conduits to gene flow, depending on the species and city in question. Thus, we speculate that dispersal ability of species and environmental heterogeneity between cities contributes to the variation exhibited in our results. However, >90% of published studies reviewed here showed an association of urbanization with genetic drift or gene flow, highlighting the strong impact of urbanization on nonadaptive evolution. It is clear that species biology and city heterogeneity obscure patterns of genetic drift and gene flow in a quantitative analysis. Thus, we suggest that future research makes comparisons of multiple cities and nonurban habitats, and takes into consideration species' natural history, environmental variation, spatial modelling and marker selection.

KEYWORDS

gene flow, genetic drift, urban facilitation model, urban fragmentation model, urbanization

1 | INTRODUCTION

Over half of the human population lives in urban areas (United Nations, 2014), and urbanization alters the landscape by replacing natural habitats with novel, human-built environments (Hobbs et al., 2006). These environments are characterized by high human population density, impervious surfaces (e.g., buildings and roads), pollution (e.g., light, air, soil and noise pollution), and habitat degradation and fragmentation (Grimm et al., 2008; Seto, Fragkias, Gueneralp, & Reilly, 2011; Seto, Gueneralp, & Hutyra, 2012). Repeated and consistent environmental modification has the capacity to influence evolutionary patterns and processes in our cities (Hendry, Gotanda, & Svensson, 2017; Johnson & Munshi-South, 2017). Increasingly, urbanization has been identified as a force that alters the strength and form of natural selection, increases mutation rates due to pollution and influences genetic drift and gene flow among populations (Box

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Box 1 Glossary

Allelic richness (Ar)
The number of alleles (*n*) observed at a microsatellite locus in a population sample.

Effective population size (N_e)
The theoretical size of an ideal population that will lose variation through genetic drift at the same rate as an actual population.

Expected heterozygosity (H_E)
The expected heterozygosity of a population based on Hardy–Weinberg genotype frequencies.

F_{IS}
The inbreeding coefficient (also designated *F*); in diploid species, the probability that an individual has two alleles at a particular locus that recently descended from a single common ancestor. Measured as $(H_E–H_O)/H_E$.

F_{ST}
The probability that two random gametes, drawn from the same subpopulation, will be identical by descent, relative to gametes taken from the entire population; developed by Wright (1951). Measured as the proportional reduction in heterozygosity as one samples individuals across populations versus within populations.

Gene flow
The transfer of genetic material from one population to another following the dispersal and subsequent reproduction of individuals, propagules or gametes.

Genetic drift
A process that changes the allele frequencies within a population stochastically from one generation to the next because of the random sampling of gametes.

Isolation by distance (IBD)
A pattern of population differentiation in which the genetic dissimilarity of populations is correlated with the geographical distance that separates them.

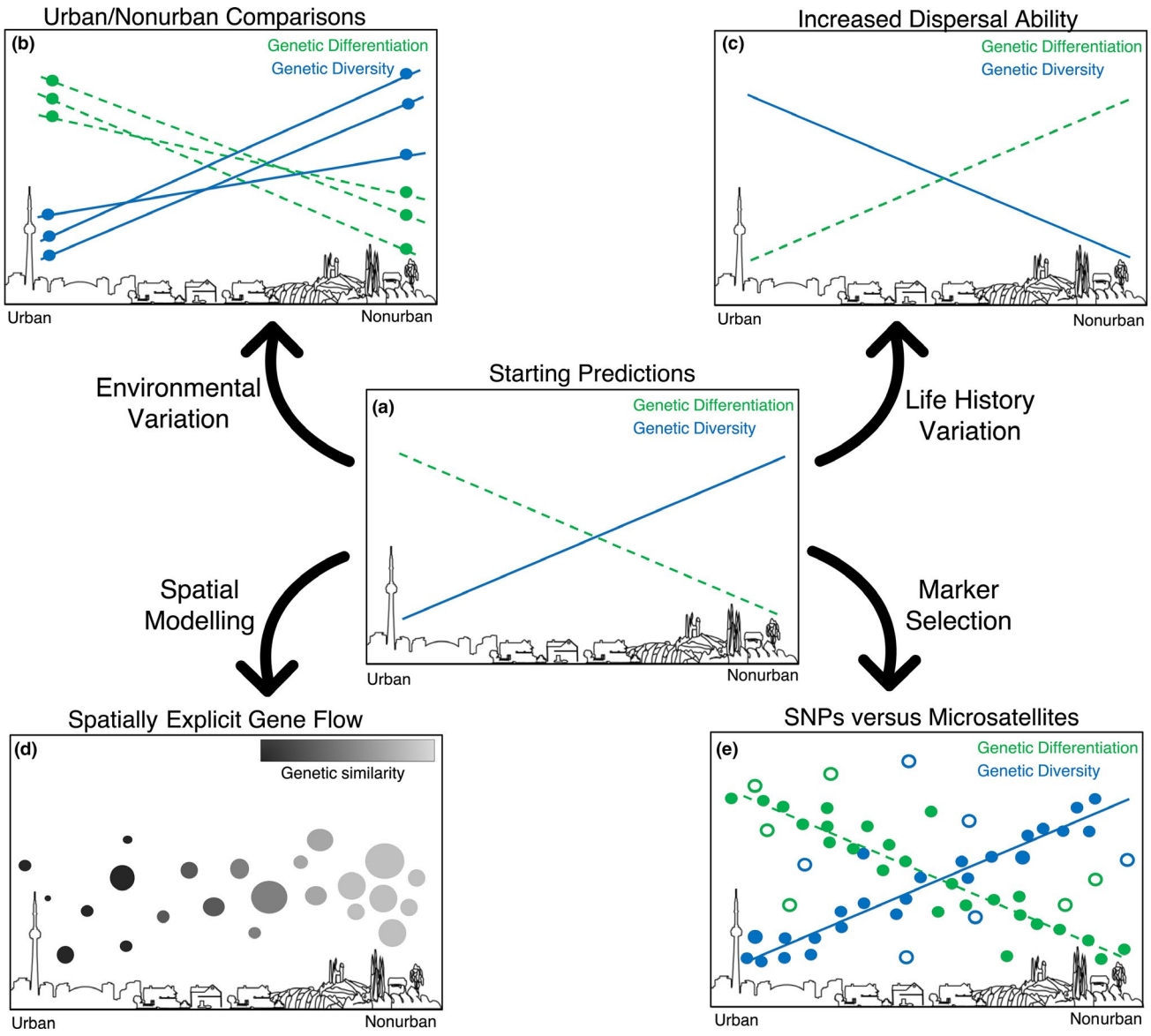
Isolation by resistance (IBR)
A pattern of population differentiation in which the genetic dissimilarity of populations is correlated with the resistance distance that separates them.

Observed Heterozygosity (H_O)
The observed heterozygosity of a population based on genotype frequencies of the population.

1; Rivkin et al., 2018). Consequently, we seek to characterize how urbanization consistently and potentially shapes patterns of evolution through a review of the literature.

Habitat fragmentation and degradation influence evolution by altering patterns of genetic drift and gene flow among populations, processes that are commonly referred to as "nonadaptive evolution." Habitat fragmentation in cities occurs when nonurban habitats are converted into living spaces for humans and related infrastructure (e.g., roads, canals and dams), reducing habitat and resources for many nonhuman organisms. This urban habitat fragmentation not only eliminates some populations but also isolates populations by impeding dispersal among them (Holderegger & Di Giulio, 2010; Storfer, Murphy, Spear, Holderegger, & Waits, 2010). This "urban fragmentation model" is the most commonly tested model in urban evolutionary studies (Crispo, Moore, Lee-Yaw, Gray, & Haller, 2011; Holderegger & Di Giulio, 2010). Urbanization-driven habitat fragmentation is expected to increase genetic drift within populations and reduce gene flow between populations. Consequently, urban habitat fragmentation is predicted to reduce genetic diversity within populations and increase genetic differentiation among populations under a standard neutral model (Box 2; e.g., Munshi-South, Zolnik, & Harris, 2016). These neutral processes are typically associated with

Box 2 Contrasting predictions from the urban fragmentation versus facilitation model. Predictions of population genetic structure vary depending on which factors have been considered



Panel a

The initial prediction of the urban fragmentation model is that genetic diversity (blue solid line/text) will decrease within and differentiation (green dashed line/text) will increase between urban habitats. However, these predictions may change when factoring in other variables.

Panel b

Environmental variation can contribute to different responses to urbanization, where responses may be constrained in urban habitats regardless of the variation in nonurban habitats. We advocate for explicit urban/nonurban comparisons to help delineate these responses.

Panel c

Variation in life history across taxa may result in alternate predictions being made (e.g., urban facilitation model). For example, highly mobile organisms may benefit from urbanization and exhibit greater genetic diversity in, and less differentiation among, urban habitats.

Box 2 (Continued)

Panel d

Such variation can also be parsed out of the data by using spatially explicit models, such as MEMGENE (Galpern, Peres-Neto, Polfus, & Manseau, 2014). Circles of similar size and colour (light vs. dark) represent individuals with similar scores, which indicate that they are associated with similar genetic and spatial factors (i.e., in the same genetic neighbourhood). Population structure due to habitat fragmentation can be more accurately identified by considering landscape features that may contribute to fragmentation.

Panel e

The choice of molecular marker can influence the predictive ability of the data. Highly variable microsatellites (open circles) may obscure patterns in the data due to homoplasy. However, the sheer abundance of SNPs (solid circles) will reduce the influence of extreme data points, increasing the likelihood of higher confidence in resolving patterns in the data.

detrimental fitness consequences, in part due to the fragmentation and isolation of populations that leads to increased drift and elevated probability of inbreeding (Brady, 2012; Cheptou, Carrue, Rouifed, & Cantarel, 2008; Leimu, Mutikainen, Koricheva, & Fischer, 2006).

However, given variation in life history and environmental conditions, alternate outcomes may be predicted (Box 2). For organisms that are highly mobile or those that thrive in urban habitats, urbanization may facilitate gene flow among populations directly via human-mediated transport, where humans voluntarily or accidentally move organisms to new habitats (Crispo et al., 2011). For example, highly mobile organisms are expected to be less impacted by habitat fragmentation than less-mobile species (Medina, Cooke, & Ord, 2018; Rochat, Manel, Deschamps-Cottin, Widmer, & Joost, 2017). Human modification to the landscape can also indirectly increase dispersal rates between populations by creating corridors (e.g., linear parks) or removing barriers (e.g., high-traffic roads or stream impoundments) to gene flow (Blair, 1996; Crispo et al., 2011; Holderegger & Di Giulio, 2010). This scenario is known as the "urban facilitation model" (Miles, Johnson, Dyer, & Verrelli, 2018). Human-facilitated gene flow reduces isolation between populations and may increase population genetic diversity by introducing novel alleles into urban populations (Box 2). Increased dispersal can reduce the likelihood of population bottlenecks and may decrease the strength of genetic drift operating within urban populations. However, newly introduced alleles can also reduce the frequency of locally adapted genotypes (e.g., Wright, 1982). Nonetheless, if selective pressures differ between urban and nonurban populations, the increased dispersal among urban populations could lead to an "urban ecotype" evolving and sweeping across urban areas (Krtinic, Ludoski, & Milankov, 2012; Schapira & Boutsika, 2012; Yakub & Tiffin, 2017).

To test whether the urban landscape is indeed the main factor driving these differences in genetic drift and gene flow between urban and nonurban populations, many studies have used a landscape genetics approach (Beninde, Feldmeier, Veith, & Hochkirch, 2018; Delgado-Ratto et al., 2016; Thomassen et al., 2018; Wood et al., 2017; Yakub & Tiffin, 2017). This approach allows researchers to identify potential barriers or conduits to gene flow between populations and make inferences about the effects of the landscape on evolutionary processes. The standard landscape genetic null model is isolation by distance (IBD, Slatkin, 1993; Wright, 1943), which predicts that as geographic distance between populations increases, so does genetic differentiation between populations. In contrast, isolation-by-resistance (IBR, McRae, 2006) models predict that in addition to geographic distance, other factors (e.g., roads and buildings) provide varying levels of resistance to gene flow, many of which act in nonlinear and nonadditive ways. Urban features may act as either complete barriers or conduits to gene flow, or may act somewhere along the continuum from barrier to conduit (Crispo et al., 2011; de Groot et al., 2016; Holderegger & Di Giulio, 2010; LaPoint, Balkenhol, Hale, Sadler, & van der Ree, 2015; Tang, Sadanandan, & Rheindt, 2016). Although there is great heterogeneity of anthropogenic landscape features found within and between cities, it is unclear the extent to which urban fragmentation and facilitation models have been applied to landscape genetic analyses.

The effects of urbanization on evolutionary processes have been increasingly studied in the past decade (Rivkin et al., 2018), and the potential impacts of urbanization on gene flow and genetic drift have been the most frequently investigated. These studies have focused on specific taxa, geographic regions and landscape features that are each expected to provide a different view of the impacts of urbanization. However, to date, no study has taken a broad analytical approach to

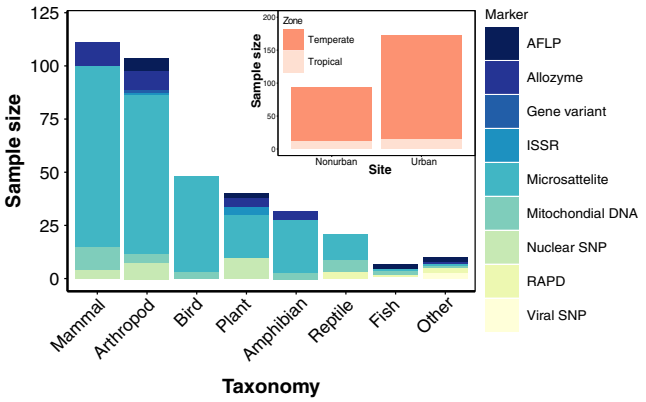


FIGURE 1 Summary of the data set used for quantitative analysis to test for differences in genetic variation. "Other" includes nematodes, annelids, gastropods, alveolates and viruses. Inset shows the number of studies performed in temperate and tropical regions for urban and nonurban habitats

examine the generality of neutral population genetic outcomes associated with urbanization. Specifically, while there is a growing trend to assume that urban fragmentation is the predominant force driving urban organismal evolution rather than urban facilitation, a rigorous examination of this assumption has not yet been attempted. Thus, with the rapid emergence of urban population genetic analyses, we sought to empirically determine how consistent and predictable the effects of urbanization are on nonadaptive evolutionary forces across diverse taxa through a review of the current literature. Here, we seek to answer two general questions: (a) How does urbanization influence genetic drift? and (b) How does urbanization influence gene flow? Following the models of urban evolution discussed above, we generated predictions about how genetic diversity and differentiation within and between populations, respectively, may evolve in response to altered genetic drift and gene flow due to urbanization. To test these predictions, we created a database of 167 research articles that empirically estimated population genetic diversity and differentiation in urban habitats (see Appendix S1 for database construction). Where available, we also extracted summary information on landscape genetic analyses within these articles to determine the role that abiotic and biotic features of urban environments play in shaping genetic variation.

2 | METHODOLOGY

We used Google Scholar and ISI Web of Science to search for studies published between 1990 and 2018 that included the following terms: "genetic drift," "genetic diversity," "landscape genetics," "population genetics" or "gene flow," together with the terms "urban" and/or "city." We define "urban" as human-modified landscapes with a high density of human dwellings, which can include towns, cities and metropolitan areas (as in United Nations, 2014). We included publications and data sets where authors identified their research as including geographic sites that were "urban" or in proximity to "urban" locales. To be as exhaustive as possible, we examined the literature cited within these research articles to identify and include studies that may have been omitted using our initial search terms. We further considered all studies listed in several recent prominent reviews (Alberti, 2015; Alberti et al., 2017; Johnson & Munshi-South, 2017; Rivkin et al., 2018). From these literature surveys, we identified 167 empirical research articles (Table S1) that met these criteria, which represents the most comprehensive data set of population genetic responses to urbanization ever assembled.

2.1 | Geographic sampling

From a broad perspective, cities are often considered replicates of each other (Alberti et al., 2017; Pickett, Cadenasso, Childers, McDonnell, & Zhou, 2016); however, they have clear differences in ecological, climatic, anthropogenic, temporal and spatial characteristics (Alberti, 2015; Grimm et al., 2008). In total, our study included species from 179 cities, with most studies conducted in North America (36%) and Europe (35%), and the remaining cities located in Asia (11%), South America (8%), Australia (7%) and Africa (3%). Across these continents, temperate zones were overrepresented (71%) among studies (Figure 1).

We identified two a priori urban-specific variables that we hypothesized would be related to how urbanization alters genetic diversity within and among populations: city area (km^2) and human population size. Because these two variables were often not included in the published studies' details, we estimated these variables from available human population data (United Nations, 2014; US Census, 2014). We found that 32 studies sampled nonurban habitats that occurred near urban areas, and because the sampling was not specific to within an urban area, we did not identify the nearby urban area size or human population for these studies. The remaining 128 studies that sampled within urban areas had urban areas that range from 0.37 to 809,000 km^2. These urban areas have human population sizes as small as 222 to as large as 21 million.

2.2 | Marker sampling

We focused our analysis on putatively neutral loci, some of which may be in linkage disequilibrium with selected regions. Adaptive evolution in the form of selective sweeps can also contribute to a loss of genetic diversity in and around genomic regions under directional selection (Hudson, Saez, & Ayala, 1997; Smith & Haigh, 1974). However, on average, the vast majority of these loci are expected to reflect nonadaptive demographic processes related to genetic drift and gene flow (Kirk & Freeland, 2011). The majority of our 167 identified studies quantified genetic diversity using nuclear microsatellite markers (N = 126), while the remaining studies used mitochondrial DNA (N = 26), allozymes (N = 14), nuclear SNPs (N = 17), AFLPs (N = 6) and ISSRs (N = 4). Of 167 studies, only 13 used multiple markers (e.g., microsatellites and mitochondrial DNA).

2.3 | Taxon sampling

The 184 species sampled in our review covered a variety of taxa: mammals (N = 62), arthropods (N = 48), amphibians (N = 21), plants (N = 20), birds (N = 15), reptiles (N = 15) and viruses (N = 3). There were 12 studies that included data from multiple species.

2.4 | Model testing

For each response variable, we built two models, a simple model that contained only urbanization as a predictor variable (Equation 1) and a more complex model that contained urbanization and covariates as predictor variables (Equation 2) (See Methods in Appendix S1 for details of response variables). We used the following linear equation models:

$$Response = \text{intercept} + \text{urbanization} + \text{study ID} + \text{error} \quad (1)$$

and

$$Response = \text{intercept} + \text{urbanization} + \text{year} + \text{kingdom} + \text{marker} + \text{number of populations} + \text{city area} + \text{city population} + \text{native} + \text{study ID} + \text{error} \quad (2)$$

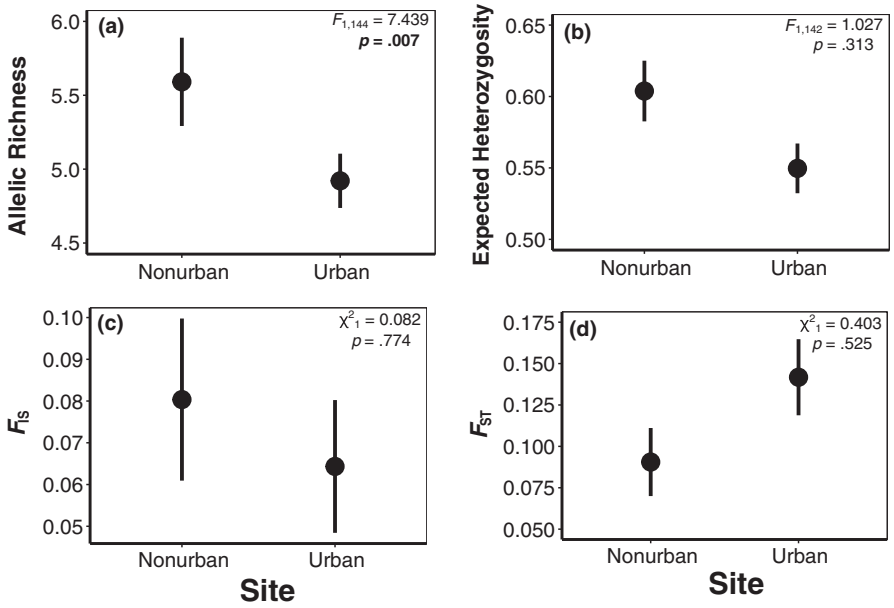


FIGURE 2 Effect of urbanization on four parameters of genetic variation (mean ± 1 *SE*) under the reduced model (Response = intercept + urbanization + study ID + error): (a) Allelic richness (Ar), (b) Expected heterozygosity (H_E), (c) F_{IS}, and (d) F_{ST}. Urban sites exhibited significantly reduced levels of Ar relative to nonurban sites. There was no significant effect of urbanization on any of the other response variables

For our complex model (2), to determine which variables best predicted the response to urbanization, we used a multi-model averaging approach. This approach identifies the best fitting model implemented, using the DREDGE function in the R package MUMIN (Barton & Barton, 2013; Methods in Appendix S1). In addition to using model averaging, we also performed a more conventional single ANOVA on each response variable. We performed the ANOVA on both the simple model (1) and the corresponding best fitting model from the DREDGE output of the complex model (2) (Methods in Appendix S1, Tables S2 and S3).

Finally, to ensure that we adequately accounted for variation in marker type and sampling design, we conducted a third analysis. In this final analysis, we estimated the effect of urbanization on genetic diversity/differentiation using only studies that estimated these parameters using microsatellites (the most numerous marker type in our database) and a paired urban–nonurban sampling design (>50% of studies per response variable) using the simple model listed above.

3 | THE INFLUENCE OF URBANIZATION ON GENETIC DRIFT: PREDICTIONS, RESULTS AND INTERPRETATION

3.1 | Predictions

In this section, we review three predictions about how genetic drift may be influenced by urbanization and discuss three examples of how these predictions apply to empirical studies.

Our first prediction is that genetic diversity within populations is unaltered by urbanization, and this prediction serves as our "null model." For example, urban and rural Tree Sparrows (*Passer montanus* Linnaeus) show no significant differences in comparisons of within-population genetic diversity in Beijing, China (Zhang, Suo, Liu, & Liang, 2013). Specifically, genetic diversity measured as allelic richness at microsatellite loci was similar within 14 urban and five rural populations (Zhang et al., 2013).

Our second prediction, following the "urban fragmentation" model, is that urbanization fragments and isolates populations, potentially increasing the strength of genetic drift. This process can lead to a loss of genetic diversity genome-wide at neutrally evolving loci within populations over time. Thus, we interpret reduced genetic diversity at neutral loci to be primarily due to drift. For example, the extent of urbanization was negatively correlated with genome-wide variation in the white-footed mouse (*Peromyscus leucopus* Rafinesque; Munshi-South et al., 2016). Specifically, genome-wide variation measured using 10,000 SNPs across 23 populations in the New York City metropolitan area was lower in urban compared with nonurban (i.e., lowest percent impervious surface) populations (Munshi-South et al., 2016). These results support the prediction that urbanization increases genetic drift in urban populations.

Our third prediction, following the "urban facilitation" model, is that urbanization may increase genetic diversity by introducing new habitats for populations, reduces isolation among them and, thus, reduces the strength of genetic drift within them. For example, genetic variation was significantly higher in urban compared with forest populations of the Great Tit (*Parus major* Linnaeus; Bjorklund, Ruiz, & Senar, 2010). Specifically, microsatellite variation was higher in the 12 urban populations compared with the nonurban forest population of Barcelona, Spain (Bjorklund et al., 2010). Although there is variance in genetic diversity within urban park populations, it was inferred that urban parks received migrants from surrounding areas. This process increased genetic diversity in the combined urban areas compared with the forest population, suggesting that genetic drift is stronger in the nonurban than the urban population.

3.2 | Results and interpretation

We extracted within-population estimates of genetic diversity, including observed (H_O: N = 161 total, N = 112 paired microsatellite

estimates) and expected heterozygosity (H_E, N = 150 total, N = 117 paired microsatellite estimates), allelic richness (Ar, N = 142 total, N = 122 paired microsatellite estimates) and the inbreeding coefficient (F_{IS}, N = 78 total, N = 53 paired microsatellite estimates) from our data set. For each study, we calculated an average for the nonurban populations as one value and the average for the urban populations as one value. We also extracted other measures of diversity (e.g., π, N = 21), but the number of studies was too few to make meaningful conclusions, and thus, we excluded these rarer metrics from the analyses. For a detailed description of the statistical analyses that were performed to test our predictions, see Methods in Appendix S1.

The majority of studies in our data set had predicted that habitat fragmentation would reduce population genetic diversity by increasing genetic drift in small and fragmented urban populations. When only urbanization was included as a predictor variable (model 1, Methods in Appendix S1), urbanization was associated with reduced genetic diversity, but this effect depended on the statistical measure of genetic diversity. Urbanization had a clear association with Ar, whereby urban populations exhibited 13% lower Ar than nonurban populations ($F_{1,143.22}$ = 7.37, p = .007; Figure 2a; Table S2). There were similar trends for other measures of genetic diversity (Figure 2 and Table S2); however, these trends were not statistically significant. The paired microsatellite data set results were consistent with those from the full data set, where urbanization was a significant predictor of Ar ($F_{1,79.89}$ = 16.25, p < .001), but not the other measures of genetic diversity.

When additional predictor variables (i.e., city area, city population, etc.) were included in the model (model 2, Methods in Appendix S1), the trends of reduced genetic diversity associated with urbanization no longer existed. Therefore, urbanization as a variable (urban vs. nonurban) was not included in the reduced model for all response variables (see Methods in Appendix S1). For Ar, the reduced model included significant effects of marker type ($F_{1,219.58}$ = 26.97, p < .001; Table S3) and number of individuals sampled ($F_{1,88.795}$ = 5.44, p = .022; Table S3). Marker type was also significantly different for H_O ($F_{3,188.46}$ = 67.51, p < .001; Table S3) and H_E ($F_{2,95.07}$ = 52.28, p < .001; Table S3). The reduced model for F_{IS} did not return any significant predictor variables (Table S3). Thus, it appears that urbanization alone is not a significant predictor of genetic diversity.

Consistent with the overall results presented above, our review of the existing literature supports all three of the predictions for the influence of urbanization on genetic drift. Overall, urbanization had a significant association with genetic diversity in 94% of studies in our data set. Specifically, the majority of studies in our data set (62%) supported the urban fragmentation model predicting reduced diversity. However, 32% of the studies supported the urban facilitation model predicting increased diversity. Finally, the null model predicting no difference between urban and nonurban sites was supported in only 6% of all studies examined. Thus, while there is evidence for reduced genetic diversity in the majority of studies, likely because urban populations experienced increased genetic drift, there is substantial counter evidence across diverse sets of taxa. Although an urban fragmentation model is often assumed in urban population genetic studies, our results surprisingly show that almost a third of published work supports the routinely ignored urban facilitation model. This fact implies that identifying universal trends in patterns of genetic diversity associated with urbanization will be challenging. Overall, our results may largely reflect the fact that patterns of genetic diversity depend heavily on species reproductive biology, dispersal ability and historical population demography (Cayuela et al., 2018). We discuss ways to address these challenges in our Section 6.

4 | THE INFLUENCE OF URBANIZATION ON GENE FLOW: PREDICTIONS, RESULTS AND INTERPRETATION

4.1 | Predictions

In this section, we review three predictions about how patterns of gene flow may be influenced by urbanization and discuss three examples of how these predictions would operate in natural systems.

Our first prediction is that genetic differentiation is unaltered in urban habitats when gene flow between populations is unaffected by urbanization, and this prediction serves as a "null model." This pattern may be a common trend among highly mobile organisms (Medina et al., 2018). For example, analyses of genome-wide SNPs in red-tailed bumblebees (*Bombus lapidarius* Linnaeus) across nine urban and nine nonurban sites in Germany found little evidence of population structure associated with urbanization and suggest panmixia (Theodorou et al., 2018). The authors argue their results are unsurprising given the mobility of the species and the large dispersal ranges of the queens and males (Lepais et al., 2010), all of which is consistent with the prediction that gene flow may be uninhibited by urbanization.

Our second prediction, following the "urban fragmentation" model, is that urbanization increases genetic differentiation by impeding gene flow among urban populations. For example, urbanization is likely responsible for reduced gene flow among seven populations of Prince of Wales feather moss (*Leptodon smithii* Hedw.; Spagnuolo, Muscariello, Terracciano, & Giordano, 2007). Specifically, ISSR markers were used to estimate differentiation among four urban and three nonurban sites in France where urban moss populations exhibited greater population genetic differentiation (F_{ST}), consistent with limited dispersal and gene flow between urban populations relative to the nonurban sites (Spagnuolo et al., 2007). The effect of urbanization on gene flow may have been exacerbated because *L. smithii* is a stationary organism with spores that are likely dispersed to unsuitable habitats (e.g., parking lots), which act as urban barriers that fragment the landscape and impede gene flow.

Our third prediction, following the "urban facilitation" model, is that urbanization decreases genetic differentiation by effectively increasing gene flow among urban populations. For example, urbanization not only facilitates gene flow among urban populations, but also among nonurban populations of the Western black widow spider (*Latrodectus hesperus* Chamberlin and Ivie; Miles, Johnson,

et al., 2018). Specifically, over 1 million genome-wide SNPs were used to estimate genetic differentiation among 10 urban and 11 nonurban populations sampled across the Western United States. Genetic differentiation (F_{ST}) was significantly lower between urban pairs than nonurban pairs of populations, and genetic connectivity (measured using population network graphs) was greater among urban populations (Miles, Johnson, et al., 2018). One possible cause of increased connectivity among urban sites is the accidental transport of *L. hesperus* individuals on vehicles and in the movement of commercial fruit crops (Miles, Johnson, et al., 2018), all of which are consistent with the prediction that urbanization may facilitate gene flow.

4.2 | Results and interpretation

We extracted estimates of F_{ST} (N = 102 total, N = 51 paired microsatellite estimates), as a measure of genetic differentiation between populations. We calculated an average for the nonurban populations as one value and the average for the urban populations as one value. Some studies used other measures of differentiation (e.g., Jost's D, N = 11). However, this number of studies was too few to make any meaningful analyses; thus, we excluded these rarer measures of differentiation for this analysis. We also recorded the number of studies that explored IBD and/or IBR in urban environments, and found that 108 studies examined the role that environmental factors play in shaping gene flow. Most studies (N = 88) tested for IBD using a Mantel test, and 57 of those studies used Mantel r^2 to report correlations. In addition to IBD, we extracted landscape resistance variables from 47 studies to examine patterns related to IBR. These data included human-created features such as percent impervious surface, road density and human population density (Alberti, 2015; Alberti et al., 2017; Grimm et al., 2008; Nowak & Greenfield, 2012), as well as natural features such as forests and rivers (Manel & Holderegger, 2013; Manel, Schwartz, Luikart, & Taberlet, 2003). For a detailed description of our statistical analyses, see Methods in Appendix S1.

Our overall analysis found no consistent association of urbanization with genetic differentiation among populations. Although F_{ST} was greater among urban populations (F_{ST} = 0.12 ± 0.16) than among nonurban populations (F_{ST} = 0.07 ± 0.06), this trend was not significant (χ^2_1 = 0.403, p = .525; Figure 2d). Similarly, there was no significant difference in F_{ST} in the paired microsatellite data set. The reduced model for F_{ST} (see Methods in Appendix S1) did not return any significant predictor variables (Table S3). Thus, while there is evidence for greater genetic differentiation between urban populations, our results show substantial variation in F_{ST} (0.03–0.68). Many studies (N = 77) calculated F_{ST} between urban populations of different cities (e.g., between Phoenix and Albuquerque, USA; Miles, Johnson, et al., 2018), whereas fewer (N = 9) calculated F_{ST} between populations within the same city (e.g., between Inwood Park and Central Park within New York City; Munshi-South et al., 2016). Consequently, the available data set allows us to make inferences about genetic differentiation between urban and nonurban populations, but does not provide sufficient power to make conclusions about genetic differentiation among urban populations within a single city.

Similar to our results for genetic diversity within populations, our results for genetic differentiation also support predictions for the influence of urbanization on gene flow. Overall, urbanization affected genetic differentiation among populations in 85% of the studies we reviewed. The majority of studies (66%) had higher genetic differentiation among urban populations and were thus consistent with the urban fragmentation model. By contrast, 19% of studies supported our prediction that differentiation was reduced among urban populations, as expected from the urban facilitation model. A further 16% of studies were consistent with the null prediction (no change in genetic differentiation). Thus, a third of the published studies support a model other than urban fragmentation in explaining patterns of genetic differentiation among urban populations. As with our analyses of genetic diversity, the effect of urbanization on genetic differentiation is likely dependent on species natural history and how species respond to urban barriers to gene flow.

Consistent with our review of the literature for the effects of urbanization on gene flow, we found no significant pattern of IBD. For the studies that sampled only urban locales (N = 22), Mantel r^2 was highly variable (0.0001–0.79) and was not statistically significant across studies (χ^2_2 = 4.96, p = .08). Nonetheless, evidence for IBD in these studies was generally very weak to moderate (r^2 = 0.12 ± 0.19), suggesting that there is little biological relevance to the statistical association. Weak signatures of IBD can be explained by both the urban fragmentation and facilitation models. The former can result in higher genetic differentiation among urban populations even at small geographic scales, whereas the latter can result in lower genetic differentiation among urban populations at all spatial scales. For example, urban fragmentation increased genetic differentiation among urban populations of striped field mice (*Apodemus agrarius* Pallas) compared with rural populations; thus, there was higher genetic differentiation at all geographic scales for urban populations, although there was no evidence of IBD (p = .87; Gortat et al., 2013). However, for urban Western black widow spiders (*L. hesperus*), genetic differentiation was overall lower for urban populations than nonurban populations as a result of urban facilitation, such that even at large geographic scales, genetic differentiation was still low between urban populations, leading to a weak signature of IBD (p = .15; Miles, Johnson, et al., 2018).

Turning to our IBR analyses, although urban land-use was identified as a barrier within many of the studies (N = 34), the strength of this trend was highly variable (r^2 = 0.009–0.42). For example, three lizard species sampled from the same urban locales experienced roads as barriers to gene flow, with the association between genetic resistance (caused by roads) and genetic distance varying from r^2 = 0.09 to 0.16 (Delaney, Riley, & Fisher, 2010). While those associations were significant, they were weak compared with genetic differentiation among red fox (*Vulpes vulpes* Linnaeus) populations, which experienced the urban landscape as a barrier to gene flow, with a significant association between genetic distance and resistance (caused by land-use) with an r^2 = 0.42 (Sacks, Brazeal, & Lewis,

2016). Six other studies found that urban land-use facilitated gene flow between populations, whereas only two studies found no statistical evidence for urban or natural barriers to gene flow (Gortat et al., 2015, 2017). Thus, as with the other measures of gene flow among urban populations, the support for urbanization as a barrier versus conduit to gene flow remains mixed.

5 | JOINT EFFECTS OF URBANIZATION ON GENETIC DRIFT AND GENE FLOW

For simplicity, we have thus far considered the effects of gene flow and genetic drift separately on urban populations, but these processes rarely act independently and it is important to consider their joint effects (Slatkin, 1985; Wright, 1931, 1951, 1969). When connectivity is decreased (i.e., gene flow is reduced), genetic drift may accelerate the loss of genetic diversity in urban populations (Epps et al., 2005). This effect may be observed if dispersal between populations regularly introduces novel alleles into the population, thereby increasing or maintaining N_e (Cayuela et al., 2018; Lande, 1992; Wright, 1969). In such cases, a loss of connectivity due to urban barriers could decrease N_e and exacerbate the effect of genetic drift on population diversity (Nei & Tajima, 1981; Wright, 1969). Similarly, genetic drift can increase population differentiation if different alleles are lost in each population (Slatkin, 1987; Young, Boyle, & Bron, 1996), an effect that may be particularly prominent in small, urban populations. Distinguishing between these two evolutionary processes is challenging and, as we note below, requires both genome-wide SNP data and landscape genetic tools to model gene flow across the urban landscape (Storfer et al., 2010).

Our review suggests that while population structure in many species can be explained by the urban habitat fragmentation model, this effect cannot be generalized across species. There were many studies that found no clear effect of urbanization on genetic diversity or were best explained by the urban facilitation model. Consequently, while patterns of genetic variation are often affected by urbanization, other factors must also be considered. These factors include but are not restricted to variation in species' natural history, sampling range, geographic heterogeneity and molecular techniques used to estimate genetic structure. We discuss ways in which these factors can be incorporated into urban evolution studies in the Section 6.

An overarching theme of our review is that the effects of urbanization on population structure of populations varies substantially among species and thus likely depends on the specific biology of each species. For example, we found that urban fragmentation can result in increased genetic drift and reduced gene flow for many organisms including both small (Gortat et al., 2013) and large mammals (Wilson, Farley, McDonough, Talbot, & Barboza, 2015), lizards (Delaney et al., 2010), amphibians (Hitchings & Beebee, 1997), fish (Mather, Hancox, & Riginos, 2015), insects (Keller, Nentwig, & Largiader, 2004) and plants (Hollingsworth & Dickson, 1997). Urban facilitation, however, resulted in reduced drift and increased gene flow in a variety of organisms including insects (Kamdem, Fouet, Gamez, & White, 2017; Vickruck & Richards, 2017), birds (Tang et al., 2016), mammals (Adams, van Heezik, Dickinson, & Robertson, 2014) and plants (Johnson, Prashad, Lavoignat, & Saini, 2018). Indeed, a wide variety of taxa also experienced no change in genetic drift and gene flow, including organisms as diverse as plants (Culley, Sbita, & Wick, 2007) and mammals (Atterby, Allnutt, MacNicoll, Jones, & Smith, 2015). These results call into question the tendency to generalize the similar pressures imposed by cities and the similar responses of even closely related species. It is important to design future research to explore how variation in traits among species and populations responds to urbanization, as we discuss below.

6 | FUTURE DIRECTIONS

6.1 | Understanding why urbanization causes differential patterns of genetic drift and gene flow across species

One of the important unresolved problems in urban evolutionary biology is why the effects of urbanization on genetic drift and gene flow vary among taxa (Box 2). For example, dispersal ability likely contributes to these differences because more mobile species (e.g., birds) are less constrained to patchy habitats than sedentary species (e.g., plants; Medina et al., 2018). Thus, studies that examine multiple taxa with varying dispersal ability within the same or similar cities may help explain patterns of variation seen among taxa. Additionally, behavioural traits may influence dispersal and success in urban environments. For example, the frequency of bold phenotypes may contribute to dispersal, such as that found in the blackbird, *Turdus merula* (Linnaeus; Evans et al., 2009; Miranda, Schielzeth, Sonntag, & Partecke, 2013), and black swan, *Cygnus atratus* (Latham; van Dongen, Robinson, Weston, Mulder, & Guay, 2015). Some urban birds have increased brain size and cognition relative to their rural counterparts (Sayol et al., 2016), that is likely contributing to increased dispersal within and between urban areas. Thus, it is clear that considering species' biology (e.g., dispersal patterns, life history, physiology) is required for generating and testing predictions of gene flow and landscape genetics. This approach is important in conducting IBD and IBR analyses, which should not exclusively focus on geographic and resistance distances, but also account for how they covary with species' characteristics across spatial scales. Comparisons of closely related species that differ in their ability to thrive in urban environments may be particularly useful.

To tease apart variation in the effects of urbanization on gene flow and genetic drift, future studies should also identify cities that differ in ecological, spatial, physical and/or socioeconomic characteristics, and determine how each of these factors affect nonadaptive evolutionary patterns. For example, the landscape factors that acted as barriers to gene flow were different across four German cities for multiple lineages of the common wall lizard (*Podarcis murali*, Laurenti; Beninde et al., 2018). Specifically, water was a significant barrier to gene flow in one city, whereas traffic volume was the

most significant barrier in another city, suggesting that both urban and natural factors can act as barriers to gene flow (Beninde et al., 2018). Thus, identifying how urban environmental heterogeneity influences patterns in genetic drift and gene flow is a critical next step for evaluating how evolution operates in cities (Rivkin et al., 2018).

A related issue is that North American and European cities in temperate areas are overrepresented in the studies we reviewed (Figure 1). Thus, our understanding of the interaction between urbanization and environmental conditions is limited by both the small geographic range and the limited climatic variation represented in the existing literature. Future studies are needed that incorporate cities in more diverse ecological, climatic, anthropogenic, temporal and spatial characteristics (Alberti, 2015; Grimm et al., 2008). For example, the Western United States has recently become highly urbanized (US Census, 2014) and has a diversity of ecoregions including deserts, grasslands, Mediterranean chaparral, montane and coastal forests (Western Ecology Division EPA, 2018). Studies in these unique ecoregions of rapid growth would provide further insight into the factors that contribute to variation in patterns of genetic drift and gene flow. For example, Western black widow spiders (*L. hesperus*) sampled in urban and nonurban populations across the Western United States experience urban-facilitated gene flow, but certain cities act as "hubs" in driving gene flow (Miles, Dyer, & Verrelli, 2018; Miles, Johnson, et al., 2018). The variation in both the ecoregions and the level of urbanization of the cities sampled likely play a large role in what drives variance in gene flow among populations (Miles, Dyer, et al., 2018). In addition, new initiatives like the Global Urban Evolution Project (GLUE; www.globalurbanevolution.comcom), which has representation from cities across six continents, may help resolve global patterns in urban evolution (Rivkin et al., 2018).

6.2 | Using a comparative approach to understand the effects of urbanization on genetic drift and gene flow

Of the studies we compiled, many did not directly compare genetic diversity or differentiation between urban and nonurban populations. Because nonurban populations serve as the baseline for determining how urbanization influences population structure, this lack of comparative sampling across urban and nonurban populations, as well as the gradient that connects them, presents a challenge to understanding the effects of urbanization on evolution. That is, urban populations may exhibit low genetic diversity, but if these patterns exist in nonurban habitats too, then these patterns may not be the result of urbanization alone (Box 2). However, for species that are only found in urban areas (e.g., bedbugs), or remnant habitat patches near urban habitats, these comparisons are impossible. Therefore, we suggest comparing population diversity of these urban species to non-anthrodependent closely related species (e.g., house sparrows and Italian sparrows; Ravinet et al., 2018). A nonmutually exclusive approach that can be applied to these urban specialists can be to focus on the features of urban environments that affect genetic drift and gene flow most strongly. For example, in anthrodependent species like bedbugs, genetic drift and gene flow can be compared between urban neighbourhoods characterized by low-density single-family housing versus high-rise housing to understand how different levels of urbanization impact neutral evolutionary processes (Booth et al., 2012; Saenz, Booth, Schal, & Vargo, 2012).

6.3 | Employing diverse genetic and genomic approaches

We suggest future studies generate SNP genotypes using high-throughput sequencing (Box 2). Such data will lead to more accurate genome-wide estimates of diversity, which will facilitate differentiating neutral and adaptive processes shaping coding and noncoding regions of the genome, and the ability to infer demographic history. As the costs for next-generation sequencing continue to decrease (Andrews, Good, Miller, Luikart, & Hohenlohe, 2016; Ekblom & Galindo, 2011), data sets will be easier and cheaper to generate that incorporate greater sample sizes for urban and nonurban sites, more cities and more taxa, all of which addresses many of the points raised above. Generating genome-wide SNPs using reduced representation methods such as ddRADseq has become a cost-effective method for use in nonmodel organisms where no reference genome is available (Peterson, Weber, Kay, Fisher, & Hoekstra, 2012). For example, with no reference genome available for the white-footed mouse, *P. leucopus*, more than 10,000 SNPs were generated by ddRADseq and then used to compare urban and rural populations as well as genome-wide patterns of variability (Munshi-South et al., 2016). These SNPs were also used to generate a robust demographic history of the populations that was previously unavailable (Harris et al., 2016). Additionally, a reference genome was generated for the burrowing owl, *Athene cunivularia*, that was then used to generate ~2 million genomic SNPs and collect demographic data for 137 owls sampled across urban and rural populations (Mueller et al., 2018). Given the breadth of SNPs collected, researchers were able to detect rare SNPs in different cities that are the result of very recent founders in each of the cities sampled (Mueller et al., 2018).

As these large data sets increase, analyses are needed that can independently tease apart the influences of genetic drift and gene flow associated with urbanization. As analytic tools are advancing rapidly, estimating N_e and patterns of complex demography in urban systems is becoming more feasible. For example, this modelling approach elucidated the historical demography that influenced the contemporary patterns of genetic variation in the brown rat, *Rattus norvegicus*, which has experienced human-mediated global expansion (Puckett & Munshi-South, 2019). Future reviews of urban evolution may be able to identify patterns of change in N_e that more directly influences patterns of genetic drift.

Currently, pairwise measures of genetic differentiation (F_{ST}) among urban and nonurban populations are commonly used. This can be problematic because there is no consensus on what is considered a significantly high value of F_{ST} (see value bins in

Frankham, Ballou, & Briscoe, 2002, 2010; Hartl & Clark, 1997; Lowe & Allendorf, 2010; Waples & Gaggiotti, 2006). For example, when F_{ST} = 0.18, this was evidence for low differentiation for mosquitoes (*Culex pipiens* Linnaeus; Asgharian et al., 2015), whereas F_{ST} = 0.18 was evidence for significantly high differentiation for Arroyo chubs (*Gila orcutti* Eigenmann & Eigenmann; Benjamin, May, O'Brien, & Finger, 2016) and ocelots (*Leopardus pardalis* Linnaeus; Janecka et al., 2011). Variation in F_{ST} reflects both the life history of an organism and its surrounding landscape, making comparisons of the magnitude of differentiation across taxa complex. Conclusions about these values alone may not reflect the impact of urban fragmentation, but rather variation related to a species' biology, such as dispersal ability.

Recent advances in landscape genetics and genomics techniques can help to overcome some of the previous pitfalls of simple measures of genetic differentiation, allowing for more accurate estimates of gene flow. Many of these new approaches measure spatially explicit patterns of genetic diversity and differentiation simultaneously, including MEMGENE (Galpern et al., 2014), EEMS (Petkova, Novembre, & Stephens, 2016) and SGD (Shirk & Cushman, 2011). For example, conditional genetic distance (CGD, Dyer & Nason, 2004) examines the covariance among all sampled subpopulations at once, and can outperform F_{ST} and other pairwise genetic distances when describing the spatial distribution of genetic variation (Dyer, Nason, & Garrick, 2010). Estimates of differential gene flow can be generated using population networks that establish the connections among populations that mirror gene flow. For example, "popgraphs" of Western black widow spiders (*L. hesperus*) showed higher connectivity among urban sites than nonurban sites and identified specific urban hubs of this connectivity that can be important for proactive pest management (Miles, Dyer, et al., 2018). Finally, including quantitative, continuous measures of urbanization will allow researchers to identify potential drivers of evolutionary change along urban gradients. For example, percent impervious surface and human density were negatively associated with reductions in genome-wide variation among *P. leucopus* (Munshi-South et al., 2016). Such measures can help inform predictions of how organisms will respond to varying levels of urbanization across multiple cities.

7 | CONCLUSIONS

Greater than 90% of published studies reviewed here reported an association of urbanization with patterns of genetic drift or gene flow, highlighting the strong impact of urbanization on nonadaptive evolution. Although our review of the literature with quantitative analyses of published urban population genetic data sets demonstrates trends towards increased genetic drift and reduced gene flow, these patterns were not significant and were not universally seen across taxa. In fact, over a third of published studies show no negative effects of urbanization on genetic diversity and differentiation, including studies supporting urban facilitation models at a much higher proportion than previously realized. How populations and species respond to urbanization clearly depends on the natural history of the taxa investigated, the number and location of cities being sampled, and the molecular techniques used to characterize population genetic structure. As urban areas continue to increase, future studies need to meet these challenges with innovation in population genetic models, taxon and population sampling, genomic data collection and landscape genetic analyses to identify the biotic and abiotic factors that contribute to the incredibly heterogeneous response that species have to urbanization.

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AUTHOR CONTRIBUTIONS

All authors contributed to the design of the review. The data set was built and analyses were conducted by L.S.M. and L.R.R. The manuscript was written by L.S.M and L.R.R., with contribution from all the authors.

DATA AVAILABILITY STATEMENT

All data and analyses have been deposited in Dryad Digital Repository https://doi.org/10.5061/dryad.02gr10f and are available from the corresponding authors upon request.

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REFERENCES

Adams, A. L., van Heezik, Y., Dickinson, K. J. M., & Robertson, B. C. (2014). Identifying eradication units in an invasive mammalian pest species. *Biological Invasions*, *16*(7), 1481–1496. https://doi.org/10.1007/s10530-013-0586-9

Alberti, M. (2015). Eco-evolutionary dynamics in an urbanizing planet. *Trends Ecology Evolution*, *30*, 114–126. https://doi.org/10.1016/j.tree.2014.11.007

Alberti, M., Correa, C., Marzluff, J. M., Hendry, A. P., Palkovacs, E. P., Gotanda, K. M., … Zhou, Y. (2017). Global urban signatures of phenotypic change in animal and plant populations. *Proceedings of the National Academy of Sciences of the USA*, *114*, 8951–8956. https://doi.org/10.1073/pnas.1606034114

Andrews, K. R., Good, J. M., Miller, M. R., Luikart, G., & Hohenlohe, P. A. (2016). Harnessing the power of RADseq for ecological and evolutionary genomics. *Nature Reviews Genetics*, *17*, 81–92. https://doi.org/10.1038/nrg.2015.28

Asgharian, H., Chang, P. L., Lysenkov, S., Scobeyeva, V. A., Reisen, W. K., & Nuzhdin, S. V. (2015). Evolutionary genomics of *Culex pipiens*: Global and local adaptations associated with climate, life-history traits and anthropogenic factors. *Proceedings of the Royal Society B Biological Sciences*, *282*, 20150728. https://doi.org/10.1098/rspb

Atterby, H., Allnutt, T. R., MacNicoll, A. D., Jones, E. P., & Smith, G. C. (2015). Population genetic structure of the red fox (*Vulpes vulpes*) in the UK. *Mammal Research*, *60*, 9–19. https://doi.org/10.1007/s13364-014-0209-6

Barton, K., & Barton, M. K. (2013). *Package 'MuMIn'*. Version, 1, 18.

Beninde, J., Feldmeier, S., Veith, M., & Hochkirch, A. (2018). Admixture of hybrid swarms of native and introduced lizards in cities is determined by the cityscape structure and invasion history. *Proceedings of the Royal Society B: Biological Sciences*, *285*(1883), 20180143. https://doi.org/10.1098/rspb.2018.0143

Benjamin, A., May, B., O'Brien, J., & Finger, A. J. (2016). Conservation genetics of an urban desert fish, the Arroyo Chub. *Transactions of the American Fisheries Society*, *145*(2), 277–286. https://doi.org/10.1080/00028487.2015.1121925c

Bjorklund, M., Ruiz, I., & Senar, J. C. (2010). Genetic differentiation in the urban habitat: The great tits (*Parus major*) of the parks of Barcelona city. *Biological Journal of the Linnean Society*, *99*, 9–19. https://doi.org/10.1111/j.1095-8312.2009.01335.x

Blair, R. B. (1996). Land use and avian species diversity along an urban gradient. *Ecological Applications*, *6*(2), 506–519. https://doi.org/10.2307/2269387

Booth, W., Saenz, V. L., Santangelo, R. G., Wang, C., Schal, C., & Vargo, E. L. (2012). Molecular markers reveal infestation dynamics of the bed bug (Hemiptera: Cimicidae) within apartment buildings. *Journal of Medical Entomology*, *49*, 535–546. https://doi.org/10.1603/ME11256

Brady, S. P. (2012). Road to evolution? Local adaptation to road adjacency in an amphibian (*Ambystoma maculatum*). *Scientific Reports*, *2*, 235. https://doi.org/10.1038/srep00235

Cayuela, H., Rougemont, Q., Prunier, J. G., Moore, J. S., Clobert, J., Besnard, A., & Bernatchez, L. (2018). Demographic and genetic approaches to study dispersal in wild animal populations: A methodological review. *Molecular Ecology*, *27*(20), 3976–4010. https://doi.org/10.1111/mec.14848

Cheptou, P. O., Carrue, O., Rouifed, S., & Cantarel, A. (2008). Rapid evolution of seed dispersal in an urban environment in the weed *Crepis sancta*. *Proceedings of the National Academy of Sciences of the USA*, *105*, 3796–3799. https://doi.org/10.1073/pnas.0708446105

Crispo, E., Moore, J., Lee-Yaw, J. A., Gray, S. M., & Haller, B. C. (2011). Broken barriers: Human-induced changes to gene flow and introgression in animals. *BioEssays*, *33*(7), 508–518. https://doi.org/10.1002/bies.201000154

Culley, T. M., Sbita, S. J., & Wick, A. (2007). Population genetic effects of urban habitat fragmentation in the perennial herb *Viola pubescens* (violaceae) using ISSR markers. *Annals of Botany*, *100*(1), 91–100. https://doi.org/10.1093/aob/mcm077

de Groot, G. A., Hofmeester, T. R., La Haye, M., Jansman, H. A. H., Perez-Haro, M., & Koelewijn, H. P. (2016). Hidden dispersal in an urban world: Genetic analysis reveals occasional long-distance dispersal and limited spatial substructure among Dutch pine martens. *Conservation Genetics*, *17*, 111–123. https://doi.org/10.1007/s10592-015-0765-6

Delaney, K. S., Riley, S. P. D., & Fisher, R. N. (2010). A rapid, strong, and convergent genetic response to urban habitat fragmentation in four divergent and widespread vertebrates. *PLoS ONE*, *5*(9), e12767. https://doi.org/10.1371/journal.pone.0012767

Delgado-Ratto, C., Gamboa, D., Soto-Calle, V. E., Van den Eede, P., Torres, E., Sánchez-Martínez, L., … D'Alessandro, U. (2016). Population genetics of *Plasmodium vivax* in the Peruvian Amazon. *PLOS Neglected Tropical Diseases*, *10*(1), e0004376. https://doi.org/10.1371/journal.pntd.0004376

Dyer, R. J., & Nason, J. D. (2004). Population graphs: The graph theoretic shape of genetic structure. *Molecular Ecology*, *13*(7), 1713–1727. https://doi.org/10.1111/j.1365-294X.2004.02177.x

Dyer, R. J., Nason, J. D., & Garrick, R. C. (2010). Landscape modelling of gene flow: Improved power using conditional genetic distance derived from the topology of population networks. *Molecular Ecology*, *19*(17), 3746–3759. https://doi.org/10.1111/j.1365-294X.2010.04748.x

Ekblom, R., & Galindo, J. (2011). Applications of next generation sequencing in molecular ecology of non-model organisms. *Heredity*, *107*, 1–15. https://doi.org/10.1038/hdy.2010.152

Epps, C. W., Palsbøll, P. J., Wehausen, J. D., Roderick, G. K., Ramey, R. R., & McCullough, D. R. (2005). Highways block gene flow and cause a rapid decline in genetic diversity of desert bighorn sheep. *Ecology Letters*, *8*(10), 1029–1038. https://doi.org/10.1111/j.1461-0248.2005.00804.x

Evans, K. L., Gaston, K. J., Frantz, A. C., Simeoni, M., Sharp, S. P., McGowan, A., … Hatchwell, B. J. (2009). Independent colonization of multiple urban centres by a formerly forest specialist bird species. *Proceedings of the Royal Society B: Biological Sciences*, *276*, 2403–2410. https://doi.org/10.1098/rspb.2008.1712

Frankham, R., Ballou, J. D., & Briscoe, D. A. (2002). *Introduction to conservation genetics*. Cambridge, UK: Cambridge University Press.

Frankham, R., Ballou, J. D., & Briscoe, D. A. (2010). *Introduction to conservation genetics*, (2nd ed.). Cambridge, UK: Cambridge University Press.

Galpern, P., Peres-Neto, P. R., Polfus, J., & Manseau, M. (2014). MEMGENE: Spatial pattern detection in genetic distance data. *Methods in Ecology and Evolution*, *5*, 1116–1120. https://doi.org/10.1111/2041-210X.12240

Gortat, T., Rutkowski, R., Gryezynska, A., Pieniazek, A., Kozakiewicz, A., & Kozakiewiez, M. (2013). Genetic structure in urban and rural populations of *Apodemus agrarius* in Poland. *Mammalian Biology*, *78*, 171–177. https://doi.org/10.1016/j.mambio.2012.07.155

Gortat, T., Rutkowski, R., Gryezynska, A., Pieniazek, A., Kozakiewicz, A., & Kozakiewiez, M. (2015). Anthropopressure gradients and the population genetic structure of *Apodemus agrarius*. *Conservation Genetics*, *16*, 649–659. https://doi.org/10.1007/s10592-014-0690-0

Gortat, T., Rutkowski, R., Gryezynska, A., Pieniazek, A., Kozakiewicz, A., & Kozakiewiez, M. (2017). The spatial genetic structure of the yellow-necked mouse in an urban environment- a recent invader vs a closely related permanent inhabitant. *Urban Ecosystem*, *20*, 581–594. https://doi.org/10.1007/s11252-016-0620-7

Grimm, N. B., Faeth, S. H., Golubiewski, N. E., Redman, C. L., Wu, J., Bai, X., & Briggs, J. M. (2008). Global change and the ecology of cities. *Science*, *319*(5864), 756–760. https://doi.org/10.1126/science.1150195

Harris, S. E., Xue, A. T., Alvarado-Serrano, D., Boehm, J. T., Joseph, T., Hickerson, M. J., & Munshi-South, J. (2016). Urbanization shapes the demographic history of a native rodent (the white-footed mouse, *Peromyscus leucopus*) in New York City. *Biology Letters*, *12*, 20150983. https://doi.org/10.1098/rsbl.2015.0983

Hartl, D., & Clark, A. G. (1997). *Principles of population genetics*. Sunderland, UK: Sinauer Associates.

Hendry, A. P., Gotanda, K. M., & Svensson, E. I. (2017). Human influences on evolution, and the ecological and societal consequences. *Proceedings of the Royal Society B: Biological Sciences*, *372*, 20160028. https://doi.org/10.1098/rstb.2016.0028

Hitchings, S. P., & Beebee, T. J. C. (1997). Genetic substructuring as a result of barriers to gene flow in urban *Rana temporaria* (common frog) populations: Implications for biodiversity conservation. *Heredity*, *79*, 117–127. https://doi.org/10.1038/hdy.1997.134

Hobbs, R. J., Arico, S., Aronson, J., Baron, J. S., Bridgewater, P., Cramer, V. A., … Zobel, M. (2006). Novel ecosystems: Theoretical

and management aspects of the new ecological world order. *Global Ecology and Biogeography*, *15*(1), 1–7. https://doi.org/10.1111/j.1466-822X.2006.00212.x

Holderegger, R., & Di Giulio, M. (2010). The genetic effects of roads: A review of empirical evidence. *Basic and Applied Ecology*, *11*(6), 522–531. https://doi.org/10.1016/j.baae.2010.06.006

Hollingsworth, P. M., & Dickson, J. H. (1997). Genetic variation in rural and urban populations of *Epipactis helleborine* (L) Crantz (Orchidaceae) in Britain. *Botanical Journal of the Linnean Society*, *123*, 321–331. https://doi.org/10.1111/j.1095-8339.1997.tb01422.x

Hudson, R. R., Saez, A. G., & Ayala, F. J. (1997). DNA variation at the Sod locus of *Drosophila melanogaster*: An unfolding story of natural selection. *Proceedings of the National Academy of Sciences of the USA*, *94*(15), 7725–7729. https://doi.org/10.1073/pnas.94.15.7725

Janečka, J. E., Tewes, M. E., Laack, L. L., Caso, A., Grassman Jr, L. I., Haines, A. M., … Honeycutt, R. L. (2011). Reduced genetic diversity and isolation of remnant ocelot populations occupying a severely fragmented landscape in southern Texas. *Animal Conservation*, *14*(6), 608–619. https://doi.org/10.1111/j.1469-1795.2011.00475.x

Johnson, M. T. J., & Munshi-South, J. (2017). The evolution of life in urban environments. *Science*, *358*(6363), 607. https://doi.org/10.1126/science.aam8327

Johnson, M. T., Prashad, C. M., Lavoignat, M., & Saini, H. S. (2018). Contrasting the effects of natural selection, genetic drift and gene flow on urban evolution in white clover (*Trifolium repens*). *Proceedings of the Royal Society B: Biological Sciences*, *285*(1883), 20181019. https://doi.org/10.1098/rspb.2018.1019

Kamdem, C., Fouet, C., Gamez, S., & White, B. J. (2017). Pollutants and insecticides drive local adaptation in African Malaria mosquitoes. *Molecular Biology and Evolution*, *34*, 1261–1275. https://doi.org/10.1093/molbev/msx087

Keller, I., Nentwig, W., & Largiader, C. (2004). Recent habitat fragmentation due to roads can lead to significant genetic differentiation in an abundant flightless ground beetle. *Molecular Ecology*, *13*, 2983–2994. https://doi.org/10.1111/j.1365-294X.2004.02310.x

Kirk, H., & Freeland, J. R. (2011). Applications and implications of neutral versus non-neutral markers in molecular ecology. *International Journal of Molecular Sciences*, *12*(6), 3966–3988. https://doi.org/10.3390/ijms12063966

Krtinic, B., Ludoski, J., & Milankov, V. (2012). Study on siphonal measurements and usefulness in delimitation of 'rural' and 'urban' ecotypes of *Culex pipiens*. *Bulletin of Insectology*, *65*, 23.

Lande, R. (1992). Neutral theory of quantitative genetic variance in an island model with local extinction and colonization. *Evolution*, *46*(2), 381–389. https://doi.org/10.1111/j.1558-5646.1992.tb02046.x

LaPoint, S., Balkenhol, N., Hale, J., Sadler, J., & van der Ree, R. (2015). Ecological connectivity research in urban areas. *Functional Ecology*, *29*, 868–878. https://doi.org/10.1111/1365-2435.12489

Leimu, R., Mutikainen, P., Koricheva, J., & Fischer, M. (2006). How general are positive relationships between plant population size, fitness and genetic variation? *Journal of Ecology*, *94*, 942–952. https://doi.org/10.1111/j.1365-2745.2006.01150.x

Lepais, O., Darvill, B., O'Connor, S., Osborne, J. L., Sanderson, R. A., Cussans, J., … Goulson, D. (2010). Estimation of bumblebee queen dispersal distances using sibship reconstruction method. *Molecular Ecology*, *19*, 819–831. https://doi.org/10.1111/j.1365-294X.2009.04500.x

Lowe, W. H., & Allendorf, F. W. (2010). What can genetics tell us about population connectivity? *Molecular Ecology*, *19*(23), 5320. https://doi.org/10.1111/j.1365-294X.2010.04688.x

Manel, S., & Holderegger, R. (2013). Ten years of landscape genetics. *Trends in Ecology & Evolution*, *28*(10), 614–621. https://doi.org/10.1016/j.tree.2013.05.012

Manel, S., Schwartz, M. K., Luikart, G., & Taberlet, P. (2003). Landscape genetics: Combining landscape ecology and population genetics. *Trends in Ecology & Evolution*, *18*(4), 189–197. https://doi.org/10.1016/S0169-5347(03)00008-9

Mather, A., Hancox, D., & Riginos, C. (2015). Urban development explains reduced genetic diversity in a narrow range endemic freshwater fish. *Conservation Genetics*, *16*, 625–634. https://doi.org/10.1007/s10592-014-0688-7

McRae, B. H. (2006). Isolation by resistance. *Evolution*, *60*, 1551–1561. https://doi.org/10.1111/j.0014-3820.2006.tb00500.x

Medina, I., Cooke, G. M., & Ord, T. J. (2018). Walk, swim or fly? Locomotor mode predicts genetic differentiation in vertebrates. *Ecology Letters*, *21*(5), 638–645. https://doi.org/10.1111/ele.12930

Miles, L. S., Dyer, R. J., & Verrelli, B. C. (2018). Urban hubs of connectivity: Contrasting patterns of gene flow within and among cities in the western black widow spider. *Proceedings of the Royal Society B: Biological Sciences*, *285*(1884), 20181224. https://doi.org/10.1098/rspb.2018.1224

Miles, L. S., Johnson, J. C., Dyer, R. J., & Verrelli, B. C. (2018). Urbanization as a facilitator of gene flow in a human health pest. *Molecular Ecology*, *27*(16), 3219–3230. https://doi.org/10.1111/mec.14783

Miranda, A. C., Schielzeth, H., Sonntag, T., & Partecke, J. (2013). Urbanization and its effects on personality traits: A result of microevolution or phenotypic plasticity? *Global Change Biology*, *19*, 2634–2644. https://doi.org/10.1111/gcb.12258

Mueller, J. C., Kuhl, H., Boerno, S., Tella, J. L., Carrete, M., & Kempenaers, B. (2018). Evolution of genomic variation in the burrowing owl in response to recent colonization of urban areas. *Proceedings of the Royal Society B: Biological Sciences*, *285*(1878), 20180206. https://doi.org/10.1098/rspb.2018.0206

Munshi-South, J., Zolnik, C. P., & Harris, S. E. (2016). Population genomics of the Anthropocene: Urbanization is negatively associated with genome-wide variation in white-footed mice populations. *Evolutionary Applications*, *9*, 546–564. https://doi.org/10.1111/eva.12357

Nei, M., & Tajima, F. (1981). Genetic drift and estimation of effective population size. *Genetics*, *98*(3), 625–640.

Nowak, D. J., & Greenfield, E. J. (2012). Tree and impervious cover change in U.S. cities. *Urban Forestry & Urban Greening*, *11*, 21–30. https://doi.org/10.1016/j.landurbplan.2012.04.005

Peterson, B. K., Weber, J. N., Kay, E. H., Fisher, H. S., & Hoekstra, H. E. (2012). Double digest RADseq: An inexpensive method for de novo SNP discovery and genotyping in model and non-model species. *PLoS ONE*, *7*(5), e37135. https://doi.org/10.1371/journal.pone.0037135

Petkova, D., Novembre, J., & Stephens, M. (2016). Visualizing spatial population structure with estimated effective migration surfaces. *Nature Genetics*, *48*(1), 94–100. https://doi.org/10.1038/ng.3464

Pickett, S. T. A., Cadenasso, M. L., Childers, D. L., McDonnell, M. J., & Zhou, W. (2016). Evolution and future of urban ecological science: Ecology in, of, and for the city. *Ecosystem Health and Sustainability*, *2*, e01229. https://doi.org/10.1002/ehs2.1229

Puckett, E. E., & Munshi-South, J. (2019). Brown rat demography reveals pre-commensal structure in eastern Asia prior to expansion into Southeast Asia. *Genome Research*, *29*, 762–770. https://doi.org/10.1101/gr.235754.118

Ravinet, M., Elgvin, T. O., Trier, C., Aliabadian, M., Gavrilov, A., & Sætre, G. P. (2018). Signatures of human-commensalism in the house sparrow genome. *Proceedings of the Royal Society B: Biological Sciences*, *285*(1884), 20181246. https://doi.org/10.1098/rspb.2018.1246

Rivkin, L. R., Santangelo, J. S., Alberti, M., Aronson, M. F. J., de Keyzer, C. W., Diamond, S. E., … Johnson, M. T. J. (2018). A roadmap for urban evolutionary ecology. *Evolutionary Applications*, *12*(3), 384–398. https://doi.org/10.1111/eva.12734

Rochat, E., Manel, S., Deschamps-Cottin, M., Widmer, I., & Joost, S. (2017). Persistence of butterfly populations in fragmented habitats along urban density gradients: Motility helps. *Heredity*, *119*, 328–338. https://doi.org/10.1038/hdy.2017.40

Sacks, B. N., Brazeal, J. L., & Lewis, J. C. (2016). Landscape genetics of the nonnative red fox of California. *Ecology and Evolution*, *6*(14), 4775–4791. https://doi.org/10.1002/ece3.2229

Saenz, V., Booth, W., Schal, C., & Vargo, E. L. (2012). Genetic analysis of bed bug (*Cimex lectularius*) populations reveals small propagule size within individual infestations but high genetic diversity across infestations from the eastern U.S. *Journal of Medical Entomology*, *49*, 535–546. https://doi.org/10.1603/ME11202

Sayol, F., Maspons, J., Lapiedra, O., Iwaniuk, A. N., Székely, T., & Sol, D. (2016). Environmental variation and the evolution of large brains in birds. *Nature Communications*, *7*, 13971. https://doi.org/10.1038/ncomms13971

Schapira, A., & Boutsika, K. (2012). Malaria ecotypes and stratification. *Advances in Parasitology*, *78*, 97. https://doi.org/10.1016/B978-0-12-394303-3.00001-3

Seto, K. C., Fragkias, M., Gueneralp, B., & Reilly, M. K. (2011). A meta-analysis of global urban land expansion. *PLoS ONE*, *6*(8), e23777. https://doi.org/10.1371/journal.pone.0023777

Seto, K. C., Gueneralp, B., & Hutyra, L. R. (2012). Global forecasts of urban expansion to 2030 and direct impacts on biodiversity and carbon pools. *Proceedings of the National Academy of Sciences of the USA*, *109*(40), 16083–16088. https://doi.org/10.1073/pnas.1211658109

Shirk, A. J., & Cushman, S. A. (2011). sGD: Software for estimating spatially explicit indices of genetic diversity. *Molecular Ecology Resources*, *11*, 922–934. https://doi.org/10.1111/j.1755-0998.2011.03035.x

Slatkin, M. (1985). Gene flow in natural populations. *Annual Reviews in Ecology, Evolution, and Systematics*, *16*, 393–430. https://doi.org/10.1146/annurev.es.16.110185.002141

Slatkin, M. (1987). Gene flow and the geographic structure of natural populations. *Science*, *236*(4803), 787–792. https://doi.org/10.1126/science.3576198

Slatkin, M. (1993). Isolation by distance in equilibrium and non-equilibrium populations. *Evolution*, *47*, 264. https://doi.org/10.1111/j.1558-5646.1993.tb01215.x

Smith, J. M., & Haigh, J. (1974). The hitch-hiking effect of a favourable gene. *Genetics Research*, *23*(1), 23–35. https://doi.org/10.1017/S0016672300014634

Spagnuolo, V., Muscariello, L., Terracciano, S., & Giordano, S. (2007). Molecular biodiversity in the moss *Leptodon smithii* (Neckeraceae) in relation to habitat disturbance and fragmentation. *Journal of Plant Research*, *120*, 595–604. https://doi.org/10.1007/s10265-007-0097-9

Storfer, A., Murphy, M. A., Spear, S. F., Holderegger, R., & Waits, L. P. (2010). Landscape genetics: Where are we now? *Molecular Ecology*, *19*(17), 3496–3514. https://doi.org/10.1111/j.1365-294X.2010.04691.x

Tang, G. S. Y., Sadanandan, K. R., & Rheindt, F. E. (2016). Population genetics of the olive-winged bulbul (*Pycnonotus plumosus*) in a tropical urban fragmented landscape. *Ecology and Evolution*, *6*(1), 78–90. https://doi.org/10.1002/ece3.1832

Theodorou, P., Radzevičiūtė, R., Kahnt, B., Soro, A., Grosse, I., & Paxton, R. J. (2018). Genome-wide single nucleotide polymorphism scan suggests adaptation to urbanization in an important pollinator, the red-tailed bumblebee (*Bombus lapidarius L.*). *Proceedings of the Royal Society B: Biological Sciences*, *285*(1877), 20172806. https://doi.org/10.1098/rspb.2017.2806

Thomassen, H. A., Harrigan, R. J., Semple Delaney, K., Riley, S. P. D., Serieys, L. E. K., Pease, K., … Smith, T. B. (2018). Determining the drivers of population structure in a highly urbanized landscape to inform conservation planning. *Conservation Biology*, *32*(1), 148–158. https://doi.org/10.1111/cobi.12969

United Nations (2014). *United Nations Department of Economic and Social Affairs, Population Division, World Urbanization Prospects: The 2014 Revision, (ST/ESA/SER.A/366).*

US Census (2014). Retrieved from http://www.census.gov/2014census/

van Dongen, W. F., Robinson, R. W., Weston, M. A., Mulder, R. A., & Guay, P. J. (2015). Variation at the DRD4 locus is associated with wariness and local site selection in urban black swans. *BMC Evolutionary Biology*, *15*(1), 253. https://doi.org/10.1186/s12862-015-0533-8

Vickruck, J. L., & Richards, M. H. (2017). Nesting habits influence population genetic structure of a bee living in anthropogenic disturbance. *Molecular Ecology*, *26*, 2674–2686. https://doi.org/10.1111/mec.14064

Waples, R. S., & Gaggiotti, O. (2006). INVITED REVIEW: What is a population? An empirical evaluation of some genetic methods for identifying the number of gene pools and their degree of connectivity. *Molecular Ecology*, *15*(6), 1419–1439. https://doi.org/10.1111/j.1365-294X.2006.02890.x

Western Ecology Division EPA (2018). Retrieved from https://www.epa.gov/aboutepa/about-western-ecology-division-wed-epas-national-health-and-environmental-effects-research.

Wilson, R. E., Farley, S. D., McDonough, T. J., Talbot, S. L., & Barboza, P. S. (2015). A genetic discontinuity in moose (*Alces alces*) in Alaska corresponds with fenced transportation infrastructure. *Conservation Genetics*, *16*, 791–800. https://doi.org/10.1007/s10592-015-0700-x

Wood, D. A., Bui, T. V. D., Overton, C. T., Vandergast, A. G., Casazza, M. L., Hull, J. M., & Takekawa, J. Y. (2017). A century of landscape disturbance and urbanization of the San Francisco Bay region affects the present-day genetic diversity of the California Ridgway's rail (*Rallus obsoletus obsoletus*). *Conservation Genetics*, *18*(1), 131–146. https://doi.org/10.1007/s10592-016-0888-4

Wright, S. (1931). Evolution in Mendelian populations. *Genetics*, *16*, 97–159.

Wright, S. W. (1943). Isolation by distance. *Genetics*, *28*, 114–138.

Wright, S. (1951). The genetical structure of populations. *Annals of Eugenics*, *15*, 323–354. https://doi.org/10.1111/j.1469-1809.1949.tb02451.x

Wright, S. (1969). *Evolution and the genetics of populations: The theory of gene frequencies* (Vol. *2*). Chicago, IL: Univ. Chicago Press.

Wright, S. W. (1982). The shifting balance theory and macroevolution. *Annual Review of Genetics*, *16*, 1. https://doi.org/10.1146/annurev.ge.16.120182.000245

Yakub, M., & Tiffin, P. (2017). Living in the city: Urban environments shape the evolution of a native annual plant. *Global Change Biology*, *23*(5), 2082–2089. https://doi.org/10.1111/gcb.13528

Young, A., Boyle, T., & Bron, T. (1996). The population genetic consequences of habitat fragmentation for plants. *Trends in Ecology & Evolution*, *11*(10), 413–418. https://doi.org/10.1016/0169-5347(96)10045-8

Zhang, S., Suo, M., Liu, S., & Liang, W. (2013). Do major roads reduce gene flow in urban bird populations? *PLoS ONE*, *8*(10), e77026. https://doi.org/10.1371/journal.pone.0077026

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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